BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Abstraction and Decomposition

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does lossy compression do?

2. What does RAM hold while a program runs?
   A) Active instructions and data
   B) Only permanent file backups
   C) Only startup firmware
   D) Only printed documents

3. "Sample rate is the number of bits used for each sample." Correct this. Explain why.

4. Two recordings have the same sample rate and length but different bit depths. "Bit depth is samples per second." Correct this. How does bit depth affect detail and size?

Today's objectives

Recall the essential ideas of abstraction and decomposition. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Decomposition
Breaking a problem into smaller parts.

Abstraction
Removing irrelevant details while retaining essential ones.

Algorithmic thinking
Developing a precise step-by-step solution.

Subproblem
Smaller task that contributes to the overall solution.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Decomposition
Break a complex problem into smaller connected subproblems, each of which can be solved and tested.

Abstraction
Keep details relevant to the task and ignore irrelevant features.

Algorithmic thinking
Describe a clear sequence of unambiguous steps for solving each subproblem.

Build and test
Combine tested parts and check that the complete solution meets the original requirements.

I Do - Worked model

A school wants a library loan system. Demonstrate decomposition and abstraction for this task. [6 marks]

- Describe the main problem
- Students must search for and borrow books, while staff track returns.
- Decompose into search catalogue, check availability, record loan and record return.
- Abstract away cover colour or shelf decorations; retain book ID, borrower ID and due date. [6]

We Do - Guided practice

1. Decompose an online ticket-booking system into three tasks. [3]

2. Explain two pieces of information to retain in a parcel-tracking system and one to ignore. [3]

3. Explain why decomposition makes a complex program easier to develop. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete this reusable function for the number of raw bitmap bits.

```
function imageBits(width,height,depth)
return ...
endfunction
print(imageBits(10,4,8))
```

You Do - Independent exam questions

Q2. Define abstraction in computational thinking. [2]

Q3. A school creates an attendance app. Suggest THREE suitable subproblems. [3]

Q4. Explain how decomposition and abstraction differ. [4]

Exit Ticket - True or False?

A trace table can help check how values change during a program.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It discards some information, so the exact original cannot necessarily be restored.

2. A - Active instructions and data

3. Sample rate is samples per second. At 8,000 Hz, 8,000 samples are taken each second. A higher rate can capture more detail but uses more space.

4. Bit depth is bits per sample. More bits allow more amplitude levels but use more space. Longer duration adds samples; sample rate and bit depth control detail.

I Do - Explain the model

Describe the main problem

Students must search for and borrow books, while staff track returns.

Decompose into search catalogue, check availability, record loan and record return.

Abstract away cover colour or shelf decorations; retain book ID, borrower ID and due date. [6]

We Do - Model answers

1. Display available tickets [1], accept and validate a booking [1], record payment or issue confirmation [1].

2. Retain parcel ID [1] and delivery status/location [1]; ignore driver's favourite music [1].

3. Smaller modules are easier to understand [1] and assign to developers [1]; individual parts can be tested [1], making errors easier to isolate [1].

Code First - Answer

```
function imageBits(width,height,depth)
return width * height * depth
endfunction
print(imageBits(10,4,8))
```

Expected output: 320

You Do - Self-mark

Q2: Focusing on relevant detail [1] and ignoring unnecessary information [1].

Q3: Accept/register pupil status [1], store attendance record [1], calculate/generate an attendance report [1]. Accept valid equivalents.

Q4: Decomposition splits a large problem into smaller parts [1] to manage complexity [1]. Abstraction hides/removes irrelevant detail [1] to focus on what matters [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.